Alexandra Gabriela Culea

+40 763 127 436 | culeaalexandra71@gmail.com | linkedin.com/in/alexandra-culea | github.com/AlexandraCulea

Education

University POLITEHNICA of Bucharest — Bucharest, Romania
Master's Degree in Advanced Techniques for Digital Imaging — *2026 – 2028*

University POLITEHNICA of Bucharest — Bucharest, Romania
Faculty of Electronics, Telecommunications and Information Technology — *2022 – 2026*

- **Specialization:** Computers and Information Technology (CTI) - INF

"Mihai Viteazul" National College — Bucharest, Romania
Mathematics and Informatics Profile — *2018 – 2022*

Experience

Cyber Security Intern — June 2025 – September 2025
Smarttech247 — *Bucharest*

- Gained hands-on experience in threat analysis, risk management, and SIEM-based threat detection using Splunk.

Projects

Heart Disease Prediction System | *Python, XGBoost, Streamlit, PostgreSQL, SQLAlchemy, Pandas* 2026

- Developed an end-to-end predictive web application for cardiovascular risk assessment.
- Trained and benchmarked multiple classifiers, XGBoost, Random Forest, alongside Logistic Regression as a baseline, saving the best-performing model.
- Integrated a robust backend with PostgreSQL and SQLAlchemy for secure data management and prediction tracking.

Multimodal Body Posture Emotion Recognition | *Python, PyTorch, ResNet18, YOLOv8n-Pose, Gemma4-26B VLM* 2026

- Engineered a multimodal deep learning system for RGB-based emotion recognition via posture analysis.
- Fused visual features extracted with ResNet18, spatial skeleton keypoints from YOLOv8n-Pose and textual descriptors from Gemma4-26B VLM.
- Evaluated and benchmarked independent feature branches alongside multi-modal fusion configurations to optimize classification performance.

Medical Clinic Management System | *Java, Spring Boot, RESTful API, MySQL, Docker, Thymeleaf* 2026

- Developed a full-stack MVC web application for clinic management.
- Implemented role-based access control with Spring Security and BCrypt, enforcing clinical workflows from visits to consultations alongside full CRUD operations.
- Built dual presentation layers featuring dynamic Thymeleaf and Bootstrap UI templates, as well as secured RESTful API endpoints tested via Postman.

Car Service RESTful API | *Node.js, Express, TypeScript, RESTful API, lowdb* 2025

- Developed a backend architecture for an auto service management system using Node.js, Express, and TypeScript.
- Implemented full CRUD operations for clients, vehicles, appointments, and service histories with lowdb persistence.

Technical Skills

Languages: Java, Python, C/C++, TypeScript, SQL (PostgreSQL, MySQL), JavaScript, HTML/CSS
Frameworks: React, Node.js, Flask, Spring Boot, Express, PyTorch, RESTful API
Developer Tools: Git, Docker, VS Code, PyCharm, IntelliJ, Postman, Splunk, Azure (Fundamentals)
Libraries: pandas, NumPy, Matplotlib, Scikit-Learn, XGBoost, SQLAlchemy, OpenCV